

Hanlong Liu

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computational Science & Engineering

Master of Science in Electrical and Computer Engineering

GPA: 4.0/4.0

Expected Graduation: 05/2027

The University of Michigan, Ann Arbor

Ann Arbor, MI

Bachelor of Science in Engineering (Computer Science Engineering)

08/2022 - 05/2025

Minor in Entrepreneurship (Ross School of Business)

GPA: 3.73/4.0

PATENTS

- Jain, D., **Liu, H.** *Earphone and Method of Use*. US Patent App. 63/706, 317 filed Oct 2024
- Developed a multi-modal earphone architecture integrating algorithmic calibration and real-time signal-processing pipelines.
- Hanlong Liu**, *A Kind of armature driver Five Unit Hi-Fi Earphone*, ZL 2019 2 0331833.4
- Designed a coordinated audio system using DSP-based tuning and multi-driver optimization algorithms to improve fidelity and stability.

INTERNSHIP

Sonic Fuse

Shenzhen, China

Founder & Product Lead

06/2020 - Present

- Led end-to-end development of custom audio hardware solutions for enterprise clients across Germany and the U.S., delivering R&D design, acoustic architecture, and manufacturable product specifications.
- Partnered with Tianqi Group (publicly listed titanium manufacturer) to design precision titanium alloy components, aligning structural design with supply chain capabilities including CNC machining and anodization processes.
- Improved production yield and reduced defect rates by optimizing tolerance control and manufacturing workflows, balancing acoustic performance with scalable production feasibility.

ByteDance – Lark (Enterprise Collaboration Platform)

Shenzhen, China

Intern, Artificial Intelligence Division

05/2023 – 08/2023

- Led deployment of an enterprise-grade LLM inference system for a large financial institution, translating business requirements (high concurrency, low latency, cost efficiency) into scalable system architecture decisions.
- Defined performance KPIs including latency, throughput, memory efficiency, and GPU utilization; drove cross-functional optimization efforts resulting in 15–30% latency reduction and 20–35% throughput improvement under high-load scenarios.
- Conducted comparative evaluation of vLLM vs. HuggingFace Transformers, producing a technical recommendation adopted by the engineering team to improve batching efficiency, KV cache management, and overall model performance (~20% improvement).
- Analyzed trade-offs between precision strategies (FP16/BF16) and parallelism configurations to balance compute cost, scalability, and reliability across A100 and 3090 environments, achieving 10–25% higher effective GPU utilization.
- Collaborated with engineering stakeholders to design batching, caching, and model-serving strategies, enabling scalable, production-ready online inference capabilities.

Development of a Patented Earphone Embedded Bone Conduction Technology

The University of Michigan, Ann Arbor

Research Intern, supervised by Prof. Dhruv Jain, Soundability Lab

03/2024- Present

- Led development of a next-generation multi-driver earphone prototype integrating DSP calibration, cross-driver compensation, and hardware-software co-design to improve audio fidelity and system reliability across diverse transducer configurations.
- Translated user-specific audiogram data into algorithmic personalization features, defining calibration logic and signal-processing requirements for individualized audio tuning.
- Designed a real-time audio processing engine (sound-source detection, adaptive noise suppression, speech enhancement), balancing latency, quality, and computational constraints for streaming use cases.
- Developed a 3-stage 3D reconstruction pipeline using structured-light data to generate anatomically accurate ear models, enabling scalable custom-fit hardware manufacturing.
- Collaborated with cross-functional teams (DSP, mechanical design, manufacturing) to deliver end-to-end prototype systems in Python/C++, aligning technical feasibility with product manufacturability.

TECHNICAL SKILLS

- Languages:** C++, Python, C
- ML / Frameworks:** PyTorch, CUDA, vLLM, HuggingFace Transformers, TensorRT, ONNX Runtime
- Systems:** Multi-GPU training/inference, Mixed Precision (FP16/BF16), Linux/Unix, Shell
- Developer Tools:** Git, Docker, NVIDIA profiling tools (nsys / nvprof)
- 3D & DSP:** Rhino 3D, AutoCAD, DSP tools, audio signal analysis
- Specialized:** Real-time inference optimization, 3D reconstruction pipeline, LLM fine-tuning, model serving & deployment